

About This Document

Statistics and probability are core to the assessment and interview (p-values, CLT, error types, distributions).

Focus areas (research-backed): Hypothesis testing, distributions and probability rules recur in both the MCQ round and the technical interview.

Descriptive Statistics

11 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. Find the mean of 2, 4, 6, 8, 10.

(A) 5

(B) 6 ✓

(C) 7

(D) 8

Answer: (B) 6

Authenticity: Community Reported

Q2. Find the median of 3, 1, 4, 1, 5, 9, 2.

(A) 2

(B) 3 ✓

(C) 4

(D) 1

Answer: (B) 3

Authenticity: Community Reported

Q3. Find the mode of 2, 3, 3, 5, 7, 3, 8.

(A) 2

(B) 3 ✓

(C) 5

(D) 7

Answer: (B) 3

Authenticity: Community Reported

Q4. Standard deviation is a measure of:

(A) Central tendency

(B) Dispersion ✓

(C) Correlation

(D) Skewness

Answer: (B) Dispersion

Authenticity: Frequently Repeated

Q5. Variance equals:

- (A) The standard deviation
- (B) The square root of the SD

(C) The square of the standard deviation ✓

- (D) The mean of deviations

Answer: (C) The square of the standard deviation

Authenticity: Community Reported

Q6. Which measure is most affected by outliers?

- (A) Mode
- (B) Median

(C) Mean ✓

- (D) Range

Answer: (C) Mean

Authenticity: Frequently Repeated

Q7. In a right-skewed distribution, the usual order is:

- (A) Mean < Median < Mode

(B) Mean > Median > Mode ✓

- (C) Mean = Median = Mode

- (D) Median > Mean > Mode

Answer: (B) Mean > Median > Mode

Authenticity: Community Reported

Q8. The median corresponds to which percentile?

- (A) 25th

(B) 50th ✓

- (C) 75th

- (D) 100th

Answer: (B) 50th

Authenticity: Community Reported

Q9. The interquartile range (IQR) is:

(A) Q3 - Q1 ✓

- (B) Q3 + Q1

- (C) Max - Min

- (D) Q2 - Q1

Answer: (A) Q3 - Q1

Authenticity: Community Reported

Q10. Find the range of 4, 8, 15, 16, 23, 42.

- (A) 23

(B) 38 ✓

- (C) 42

- (D) 34

Answer: (B) 38

Authenticity: Research-Backed Company Style

Q11. The mean is heavily influenced by, but the median is robust to:

(A) Sample size

(B) Outliers ✓

(C) Units

(D) Rounding

Answer: (B) Outliers

Authenticity: Community Reported

Probability

10 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. If $P(A)=0.3$, $P(B)=0.5$, independent, then $P(A \text{ and } B)$ is:

(A) 0.8

(B) 0.15 ✓

(C) 0.2

(D) 0.5

Answer: (B) 0.15

Authenticity: Community Reported

Q2. For mutually exclusive events $P(A)=0.2$, $P(B)=0.3$, $P(A \text{ or } B)$ is:

(A) 0.06

(B) 0.5 ✓

(C) 0.1

(D) 0.25

Answer: (B) 0.5

Authenticity: Community Reported

Q3. Expected value of a single fair die roll:

(A) 3

(B) 3.5 ✓

(C) 4

(D) 2.5

Answer: (B) 3.5

Authenticity: Frequently Repeated

Q4. Two fair coins tossed; $P(\text{exactly one head})$:

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$ ✓

(C) $\frac{3}{4}$

(D) $\frac{1}{3}$

Answer: (B) $\frac{1}{2}$

Authenticity: Community Reported

Q5. If A and B are independent, $P(A | B)$ equals:

(A) $P(B)$

(B) $P(A)$ ✓

(C) $P(A \text{ and } B)$

(D) 0

Answer: (B) $P(A)$

Authenticity: Frequently Repeated

Q6. Sum of probabilities of all outcomes in a sample space:

(A) 0

(B) 0.5

(C) 1 ✓

(D) Depends

Answer: (C) 1

Authenticity: Community Reported

Q7. Bayes' theorem is used to:

(A) Compute variance

(B) Relate $P(A|B)$ to $P(B|A)$ ✓

(C) Test hypotheses

(D) Measure correlation

Answer: (B) Relate $P(A|B)$ to $P(B|A)$

Authenticity: Frequently Repeated

Q8. Probability a drawn card is a heart:

(A) $1/13$

(B) $1/4$ ✓

(C) $1/52$

(D) $1/2$

Answer: (B) $1/4$

Authenticity: Community Reported

Q9. Two coins tossed; $P(\text{at least one head})$:

(A) $1/4$

(B) $1/2$

(C) $3/4$ ✓

(D) 1

Answer: (C) $3/4$

Authenticity: Research-Backed Company Style

Q10. The conditional probability $P(A|B)$ is defined as:

(A) $P(A)P(B)$

(B) $P(A \text{ and } B)/P(B)$ ✓

(C) $P(B)/P(A)$

(D) $P(A)+P(B)$

Answer: (B) $P(A \text{ and } B)/P(B)$

Authenticity: Frequently Repeated

Inferential Statistics & Hypothesis Testing

12 practice questions. Correct option highlighted; authenticity label shown under each answer.

Q1. A p-value below the significance level suggests you should:

(A) Accept the null

(B) Reject the null hypothesis ✓

(C) Ignore the result

(D) Increase alpha

Answer: (B) Reject the null hypothesis

Authenticity: Frequently Repeated

Q2. A Type I error is:

(A) Failing to reject a false null

(B) Rejecting a true null hypothesis ✓

(C) Accepting correctly

(D) A sampling error

Answer: (B) Rejecting a true null hypothesis

Authenticity: Frequently Repeated

Q3. A Type II error is:

(A) Rejecting a true null

(B) Failing to reject a false null hypothesis ✓

(C) A measurement error

(D) A rounding error

Answer: (B) Failing to reject a false null hypothesis

Authenticity: Frequently Repeated

Q4. The Central Limit Theorem says the sampling distribution of the mean:

(A) Is always uniform

(B) Approaches normal as sample size grows ✓

(C) Has zero variance

(D) Equals the population distribution

Answer: (B) Approaches normal as sample size grows

Authenticity: Frequently Repeated

Q5. For a normal distribution, ~what percent lies within one SD of the mean?

(A) 50%

(B) 68% ✓

(C) 95%

(D) 99.7%

Answer: (B) 68%

Authenticity: Community Reported

Q6. For a normal distribution, ~what percent lies within two SD of the mean?

- (A) 68%
- (B) 90%
- (C) 95% ✓**
- (D) 99.7%

Answer: (C) 95%

Authenticity: Community Reported

Q7. Pearson's correlation coefficient ranges between:

- (A) 0 and 1
- (B) -1 and 1 ✓**
- (C) -inf and inf
- (D) 0 and 100

Answer: (B) -1 and 1

Authenticity: Frequently Repeated

Q8. A correlation coefficient of 0 indicates:

- (A) Perfect positive
- (B) Perfect negative
- (C) No linear relationship ✓**
- (D) Causation

Answer: (C) No linear relationship

Authenticity: Community Reported

Q9. As sample size increases, the standard error of the mean:

- (A) Increases
- (B) Decreases ✓**
- (C) Stays constant
- (D) Becomes negative

Answer: (B) Decreases

Authenticity: Community Reported

Q10. Which test compares the means of two groups?

- (A) Chi-square
- (B) t-test ✓**
- (C) One-group ANOVA
- (D) Correlation

Answer: (B) t-test

Authenticity: Frequently Repeated

Q11. A/B testing is essentially a form of:

- (A) Clustering
- (B) Hypothesis testing ✓**
- (C) Dimensionality reduction
- (D) Data cleaning

Answer: (B) Hypothesis testing

Authenticity: Community Reported

Q12. A positive covariance between X and Y indicates:

- (A) They move in opposite directions
- (B) They tend to move in the same direction ✓**
- (C) No relationship
- (D) X causes Y

Answer: (B) They tend to move in the same direction

Authenticity: Community Reported

Tredence

Statistics & Probability

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

Table of Contents

About This Document	1
Descriptive Statistics	1
Probability	3
Inferential Statistics & Hypothesis Testing	5
Authenticity Legend & How to Read This Document	9

Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.